

Hospital AI Agent

Use Case Specification

1. System Actors

The Hospital AI Agent system has the following main actors:

1. Hospital Administrator.
2. Doctor.
3. Nurse.
4. Receptionist.
5. Patient.
6. AI Agent.

Each actor interacts with the system based on their assigned role and permissions.

2. Administrator Use Cases

The Hospital Administrator is responsible for managing and monitoring the hospital system.

The Administrator can:

- Log in to the system.
- View the hospital dashboard.
- Manage doctors.
- Add doctors.
- Edit doctor information.
- Remove doctors.
- Manage nurses.
- Assign nurses to departments.
- Monitor hospital statistics.

- View daily appointments.
- View hospital revenue.
- Generate reports.
- Monitor doctor workload.
- Receive administrative alerts.

Administrator Use Case Flow

Administrator

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Login

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Dashboard

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Select Management Function

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Manage Doctors / Nurses / Reports / Statistics

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View Results

3. Doctor Use Cases

The Doctor interacts with the system to manage appointments and patient information.

The Doctor can:

- Log in.
- View personal schedule.
- View upcoming appointments.
- View authorized patient information.
- View authorized medical history.

- Update appointment status.
- Mark an appointment as completed.
- Add medical notes.
- Add prescriptions.
- View personal appointment statistics.

Doctor Use Case Flow

Doctor

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Login

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Doctor Dashboard

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View Schedule

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Select Appointment

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View Authorized Patient Information

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Complete Appointment

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Update Appointment Status

4. Nurse Use Cases

The Nurse uses the system to manage work assignments and shifts.

The Nurse can:

- Log in.

- View assigned department.
- View assigned doctor.
- View daily shift.
- View working hours.
- Receive shift notifications.
- Confirm shift completion.

Nurse Use Case Flow

Nurse

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Login

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Nurse Dashboard

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View Shift

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View Assigned Department

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Complete Shift

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Confirm Completion

5. Receptionist Use Cases

The Receptionist is responsible for assisting patients and managing appointments.

The Receptionist can:

- Log in.
- Register a new patient.

- Search for patients.
- Search for doctors.
- View doctor availability.
- Book appointments.
- Modify appointments.
- Cancel appointments.

Receptionist Use Case Flow

Receptionist

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Login

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Reception Dashboard

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Search or Register Patient

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Select Doctor

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Check Available Appointment

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Book Appointment

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Confirm Booking

6. Patient Use Cases

The Patient can interact with the system to manage personal appointments.

The Patient can:

- Register an account.
- Log in.
- View doctors.
- Search for doctors.
- View doctor specialties.
- Book appointments.
- Modify appointments.
- Cancel appointments.
- View appointment history.
- Receive appointment notifications.
- View authorized personal medical information.
- Interact with the AI Agent.

Patient Use Case Flow

Patient

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Login

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Patient Dashboard

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Select Service

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Book Appointment / View Appointments / Talk to AI Agent

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Receive Result

7. AI Agent Use Cases

The AI Agent is an intelligent component that assists users.

The AI Agent can:

- Understand user requests.
- Answer hospital-related questions.
- Search for doctors.
- Suggest appropriate specialties.
- Check available appointments.
- Suggest appointment times.
- Assist with appointment booking.
- Retrieve authorized information.
- Generate summaries.
- Analyze workload.
- Generate intelligent reports.

The AI Agent does not directly perform unauthorized actions.

All actions are performed through controlled system tools and permissions.

8. AI Agent Appointment Booking Use Case

Use Case Name

AI-Assisted Appointment Booking

Primary Actor

Patient

Secondary Actor

AI Agent

Preconditions

- The patient is registered or identified by the system.
- Doctor schedules are available.

Main Flow

1. The Patient sends a request to the AI Agent.
2. The AI Agent understands the request.
3. The AI Agent identifies the required specialty.
4. The AI Agent searches for available doctors.
5. The AI Agent checks available appointment slots.
6. The AI Agent suggests available options.
7. The Patient selects a preferred option.
8. The system creates the appointment.
9. The Patient receives confirmation.

Postcondition

The appointment is successfully stored in the system.

9. Appointment Management Use Case

Use Case Name

Book Appointment

Primary Actor

Patient or Receptionist

Preconditions

- Patient information exists in the system.
- Doctor schedule is available.

Main Flow

1. The user selects a medical specialty.
2. The system displays available doctors.
3. The user selects a doctor.
4. The system displays available appointment slots.

5. The user selects a date and time.
6. The system validates availability.
7. The appointment is created.
8. The system sends a confirmation.

Alternative Flow

If the selected appointment is not available:

1. The system displays alternative appointment slots.
 2. The user selects another appointment.
 3. The booking process continues.
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10. Patient Registration Use Case

Use Case Name

Register Patient

Primary Actor

Receptionist

Preconditions

The receptionist is logged into the system.

Main Flow

1. The receptionist enters patient information.
 2. The system validates the information.
 3. The system checks for duplicate records.
 4. The system creates a patient profile.
 5. The system confirms successful registration.
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11. Doctor Management Use Case

Use Case Name

Manage Doctors

Primary Actor

Administrator

Main Flow

1. The Administrator logs into the system.
 2. The Administrator opens Doctor Management.
 3. The system displays the doctor list.
 4. The Administrator can add, edit, or remove doctors.
 5. The system validates the changes.
 6. The system updates the database.
 7. The system confirms the operation.
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12. AI Agent Interaction Use Case

Use Case Name

Ask AI Agent

Primary Actor

Patient / Doctor / Administrator

Main Flow

1. The user sends a natural language request.
2. The AI Agent analyzes the request.
3. The AI Agent identifies the user's intent.
4. The AI Agent determines whether a tool is required.
5. The Agent calls the appropriate authorized tool.
6. The system returns the requested information.
7. The AI Agent provides a response.

Example:

User:

“I need an appointment with a dermatologist.”

AI Agent:

1. Identifies the specialty.
 2. Searches for dermatologists.
 3. Checks available appointments.
 4. Suggests available options.
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13. Notification Use Case

Use Case Name

Send Appointment Reminder

Primary Actor

System / AI Agent

Secondary Actor

Patient

Main Flow

1. The system checks upcoming appointments.
2. The system identifies appointments that require reminders.
3. The notification service sends a reminder.
4. The patient receives the notification.

Possible reminder times include:

- 24 hours before the appointment.
 - 1 hour before the appointment.
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14. Report Generation Use Case

Use Case Name

Generate Hospital Report

Primary Actor

Administrator

Main Flow

1. The Administrator selects the report type.
2. The system retrieves the required data.
3. The system analyzes the data.
4. The system generates statistics.
5. The report is displayed to the Administrator.

Possible reports include:

- Daily patient report.
- Monthly patient report.
- Doctor performance report.
- Department performance report.
- Revenue report.
- Appointment statistics.

15. Summary

The Hospital AI Agent system supports multiple users, each with specific responsibilities and permissions.

The main interaction areas are:

- Hospital management.
- Doctor management.
- Nurse management.
- Patient management.
- Appointment management.

- Notifications.
- Reporting.
- AI-powered assistance.

The AI Agent acts as an intelligent layer that helps users interact with the system more easily while using controlled tools and respecting user permissions.